

Program Committee Report

Business meeting

TQC 2025

Bill Fefferman (Program chair)

How were talk decisions made by the PC?

- Each submitted paper was reviewed and given a numeric score by at least 3 committee members during the “review phase”
- Papers with the highest total score were accepted
- Borderline papers were decided by the entire PC during the “discussion phase”, after all reviews were submitted

Changes from last year: Call for submissions

As with last year, there were two tracks for submissions

- A. With Proceedings track (talk + proceedings)
- B. Without Proceedings track (talk only)

Modification from previous years

Both tracks had same submission requirements --

- 1. Technical manuscript (no page limit), authors encouraged to use full version of paper – first 15 pages will be published for papers accepted to the With Proceedings track**
- 2. Extended abstract (at most 3 pages not including references)**

Submission deadlines and issues

- Submission deadlines
 - March 1, 2025 (papers in both tracks)
 - April 30th (posters)
- This was the first time TQC used a HotCRP server for all paper submissions, graciously hosted at the University of Amsterdam
- We switched from EasyChair which had been used in all prior editions of TQC
- *Issues*
 - HotCRP server had a glitch which kept some fraction of notifications from being sent out
 - It seems the problem was with the email server, which was setup with certain limits to avoid spam, keeping us from sending out large batches of emails in a short period of time
 - To fix this issue, I manually resent the notification in small batches (~100 notifications each)
 - I'm told this has been resolved for QIP, which is using the same server!

Reviewing and discussion

- **Review phase:** March 2nd to April 8th
 - Program committee members independently write reviews for their papers, with the help of subreviewers
- **Discussion phase:** April 9th to April 24th
 - Program committee shares reviews and achieve consensus
- **Notifications:** April 25th
- **The PC committee treated papers in the With and Without proceedings tracks equally**
 - (The publisher requires at least 10 papers to be published in each edition -- we had no problems achieving the required number of papers for the proceedings)

Best paper awards (selected by PC members via vote)

- **Best student paper award:** “Quantum Purity Amplification: Optimality and Efficient Algorithm”, by Zhaoyi Li, Honghao Fu, Takuya Isogawa, Caio Silva, Isaac Chuang
- **Best paper award:** "Polylog-time- and constant-space-overhead fault-tolerant quantum computation with quantum low-density parity-check codes", by Shiro Tamiya, Masato Koashi, Hayata Yamasaki

Congratulations to all authors!

Posters

- We accepted as posters both any non-accepted talk submission and poster-only submissions that were
 - A. Judged by the program committee to be correct (i.e., not clearly wrong)
 - B. in scope for the conference
 - C. in the right format

Statistics

- 375 total paper submissions (460 paper submissions in 2024)
- 3-4 reviews per submission
 - Past years used 3 reviews per paper, in most cases we were able to increase to 4 reviews, which led to more confident PC decisions
 - PC load: ~20-23 papers per PC member
- 77 out of 309 (~25%) accepted for talks in Without Proceedings track
- 12 out of 66 (~18%) accepted for talks in With Proceedings track
- Total: 89 accepts out of 375 = 23.7% accept rate

Thanks (1) - PC members and subreviewers

58 PC members in all

424 subreviewers

PC members:

- Bill Fefferman (University of Chicago, Program Chair)
- Simon Apers (Université Paris Cité)
- Aleksandrs Belovs (University of Latvia)
- Adam Bene Watts (University of Calgary)
- Michael Beverland (IBM Quantum)
- Nikolas Breuckmann (University of Bristol)
- Anne Broadbent (University of Ottawa)
- Anthony Chen (Simons Institute, Berkeley)
- Matthias Christandl (University of Copenhagen)
- Andrea Coladangelo (University of Washington)
- Alexander Dalzell (Amazon)
- Abhinav Deshpande (IBM Quantum)
- Di Fang (Duke University)
- Omar Fawzi (Inria and École Normale Supérieure de Lyon)
- Sevag Gharibian (Paderborn University)
- Daniel Grier (University of California San Diego)
- Vojtech Havlicek (IBM Quantum)
- Jiaqing Jiang (Caltech)
- Tamara Kohler (Stanford)
- Aleksander Kubica (Yale)
- Richard Kueng (Johannes Kepler University Linz)
- Srijita Kundu (University of Waterloo)
- Ludovico Lami (Scuola Normale Superiore, Pisa)
- Cécilia Lancien (Institut Fourier Grenoble and CNRS)
- Nicholas Laracuente (Indiana University Bloomington)
- Felix Leditzky (University of Illinois Urbana-Champaign)
- Troy Lee (University of Technology Sydney)
- Jiaqi Leng (Simons Institute, Berkeley)
- Jiahui Liu (Fujitsu Research)
- Saeed Mehraban (Tufts University)
- Tomoyuki Morimae (Yukawa Institute for Theoretical Physics, Kyoto University)
- Hui Khoo Ng (National University of Singapore)
- Nelly Ng (Nanyang Technological University)
- Harumichi Nishimura (Nagoya University)
- Changhun Oh (Korea Advanced Institute of Science and Technology)
- Māris Ozols (University of Amsterdam, QuSoft)
- David Pérez-García (Universidad Complutense de Madrid)
- Alexander Poremba (MIT)
- Luke Schaeffer (University of Waterloo)
- Norbert Schuch (University of Vienna)
- Thomas Schuster (Caltech)
- Christian Schaffner (University of Amsterdam, QuSoft)
- Kunal Sharma (IBM Quantum)
- Makrand Sinha (University of Illinois Urbana-Champaign)
- Graeme Smith (University of Waterloo)
- Mehdi Soleimanifar (Caltech)
- Fang Song (Portland State University)
- Daniel Stilck França (University of Copenhagen)
- Sergii Strelchuk (Oxford University)
- Marco Tomamichel (Centre for Quantum Technologies, National University of Singapore)
- Dave Touchette (Université de Sherbrooke)
- Benjamin Villalonga (Google)
- Dominic Williamson (University of Sydney)
- Freek Witteveen (University of Copenhagen and QuSoft)
- Takashi Yamakawa (NTT Social Informatics Laboratories)
- Jiayu Zhang (Zhongguancun Laboratory, Beijing)
- Ruizhe Zhang (Simons Institute, Berkeley)
- Sisi Zhou (Perimeter Institute)

Thanks (2) – Local and international organizers

Local organizers in Bengaluru

- C. M. Chandrashekar, IISc, Bengaluru
- Baladitya Suri, IISc, Bengaluru
- Navin Kashyap, IISc, Bengaluru
- Henry Sukumar, C-DAC, Bengaluru
- Krishnakumar Sabapathy, Fujitsu Research, Bengaluru

National organizers

- Anirban Pathak, IIIT Noida
- Prabha Mandayam, IIT Madras
- Sudhir Kamath, DRDO-IARCOE
- Manik Banik, SN Bose National Centre for Basic Sciences, Kolkata
- Saibal K. Pal , SAG-DRDO Delhi

International organizers

- Lída del Rio, Squids and University of Zurich
- Nuriya Nurgalieva, Squids and University of Zurich

Thanks (3) – Steering committee

- Kai-Min Chung, Academia Sinica
- Steve Flammia, AWS Center for Quantum Computing
- Min-Hsiu Hsieh, Hon Hai (Foxconn)
- Shelby Kimmel, Middlebury College
- François Le Gall, Nagoya University (chair)
- Frederic Magniez, CNRS (co-chair)
- Kae Nemoto, OIST
- Lídia del Rio, Squids and University of Zurich

Thanks (4): to everyone for attending TQC 2025!